QP CODE: 25805793

25805793

Reg No : .....

Name : .....

I M C A DEGREE EXAMINATION, DECEMBER 2025

Ninth Semester

INTEGRATED MCA

CORE - IMCA9C03 - DATA SCIENCE

2020 Admission Onwards

997155F9

Time: 3 Hours Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Explain why Big Data is important for businesses?
2. Compare prescriptive and predictive analytics in terms of decision-making.
3. Why is the p-value important in hypothesis testing?
4. What is the main goal of multivariate data analysis?
5. Why is labeled data important in supervised learning?
6. How does machine learning help in weather forecasting?
7. Explain different types of classification algorithms.
8. What is discriminent function? Explain two different types of discriminent function.
9. Is KNN a parametric or Non parametric algorithm? Justify your answer.
10. List and briefly describe any two techniques used for feature selection.
11. What is dimensionality reduction? Why it is important?
12. Explain the steps involved in performing PCA.

(10×3=30 marks)

Part B

*Answer **all** questions*

*Each question carries **9** marks*

13. a) Explain types of Big Data with example.

OR

b) How do businesses use different types of statistics to improve performance?

14. a) A call center receives an average of 10 calls per hour. What is the probability that no calls are received in a given hour?

OR

b) A factory was producing electric bulbs of average length of 2000 hours. A new manufacturing process was introduced with the hope of increasing the length of the life of bulbs. A sample of 25 bulbs produced by the new process were examined and the average length of life was found to be 2200 hours. Examine whether the average length of bulbs was increased assuming the length of lives of bulbs follow normal distribution with 0.05.

15. a) Briefly expalin different types of machine learning.

OR

b) Discuss the working of k-Means Clustering with an example.

16. a) Compare and contrast clustering and classification in data mining.

OR

b) What is Support Vector Machine and how does it find the optimal hyperplane for classification?

17. a) Define feature Selection. What are all the three methods for feature selection?

OR

b) Explain how data science is used for fraud detection and risk asssessment in detail.

(5×9=45 marks)